# Ismail Kuru, Ph.D.

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## Employment History

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|---|---|
| 2026(Feb) – | **Postdoctoral Researcher.** Aarhus University, Department of Computer Science, Center for Basic Research in Program Verification. Working with Prof. Lars Birkedal and Prof. Jean Pichon-Pharabod. |
| 2026(Oct) – | **Postdoctoral Researcher.** Center for R&D on Secure Computer Systems (CRADSEC), Research Organization of Information and Systems (ROIS), Tokyo, Japan. Working with Prof. Taro Sekiyama. |
| 2026(Apr) – 2026(Sep) | **Visiting Researcher.** National Institute of Informatics, Tokyo, Japan. Hosted by Prof. Taro Sekiyama; co-advised a summer research intern. |
| 2025(July) – 2026(Feb) | **Postdoctoral Researcher.** Drexel University, College of Computing & Informatics. |
| 2016(Oct) – 2025(June) | **Doctoral Research Fellow.** Drexel University, College of Computing & Informatics. |
| 2019(Apr) – 2019(June) | **Research Intern.** BedRock Systems. |
| 2016(Jan) – 2016(Sep) | **Senior Software Engineer.** Crytek GmbH. |
| 2014(Sep) – 2015(Feb) | **Research Assistant.** Koc University. Advisor: Dr. Serdar Tasiran. |
| 2014(Apr) – 2014(Aug) | **Research Intern.** Microsoft Research. Advisors: Dr. Matthew J. Parkinson, Ben Hall and Serdar Tasiran. |
| 2013(Sep) – 2014(Mar) | **Visiting Research Student.** University of Washington. Advisor: Dr. Dan Grossman. |
| 2012(Sep) – 2013(Sep) | **Research Assistant.** Koc University. |
| 2011(Mar) – 2011(Sep) | **Software Developer.** GNU Compiler Collection, Google Summer of Code 2011. Advisor: Dr. Albert Cohen. |

## Education

2018 – 2025 **Ph.D., Drexel University, U.S.A** in Software Reliability.
Thesis: *Modal Abstractions for Operating System Kernels*
Advisor: *Dr. Colin S. Gordon*
Funding: NSF Career Project (*link*)

2016 – 2018 **M.Sc. Computer Science, Drexel University, U.S.A.**
Internships: BedRock Systems
Coursework Completed

2015 **M.Sc. Computer Science, Koc University, Turkey** in Software Verification.
Thesis title: *Static Methods for Checking Correctness of Programs on Relaxed Memory Systems.*
Advisor: *Dr. Serdar Tasiran*
Scholarship: 1 of 18 Microsoft Research Europe PhD Scholarships (*link*)
Internships: University of Washington Visit and MSR Cambridge
Coursework Completed in Technical University of Munich, Germany.

## Teaching and Mentoring

Drexel
- **Teaching Assistant**, CS 360 Programming Languages; led discussion sections.
- **Teaching Assistant**, SE 311 Software Engineering.

Koc
- **Teaching Assistant**, Software Engineering, Koc University.

2026
- **Research Advisor**, Keisuke Nishimura (research intern, National Institute of Informatics, summer 2026), program logic for inline assembly in Iris; co-advised with Taro Sekiyama through weekly meetings.

2021 –
- **Long-Term Mentor**, SIGPLAN-M (ACM SIGPLAN mentoring program), mentoring graduate and senior undergraduate students in programming languages research.

## Skills

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|---|---|
| Languages | Strong reading, writing and speaking in English and Turkish. Intermediate German. |
| SMT Based Verification | VCC, Boogie, QED. |
| Model Checking | NuSMV. |
| Type Theory Based Verification | Rocq (formerly Coq), Iris. |

## Miscellaneous Experience

### Awards and Achievements

2018
- **Scholarship**. DeepSpec Summer School Scholarship Princeton U.S.A, 2018.

2017
- **Scholarship**. PLMW Scholarship for ICFP'17 Oxford U.K, 2017.

2014
- **Travel Grant**. Microsoft Research Visitor Grants(Multiple) Hosted by Dr. Matthew Parkinson, 2014.

2011–2014
- **Graduate School Scholarship**. Received one of 18 Microsoft Research Europe PhD Scholarship awards, 2011-2014.

2012
- **Travel Grant**. from Koc University for VCLA Winter School Vienna, 2012.

2011
- **Travel Grant**. Inria-Paris Visitor Grants(Multiple) Hosted by Dr. Albert Cohen, 2011.

### Summer/Winter Schools

2018
- DeepSpec Summer School Princeton U.S.A, 2018.

2017
- PLMW for ICFP'17 Oxford U.K, 2017.
- OPLSS Summer School Oregon U.S.A, 2017.

2012
- ACACES Eighth International Summer School Italy, 2012.
- Microsoft Research Ph.D. Summer School Cambridge U.K, 2012.
- Vienna Center for Logic and Algorithms Winter School on Verification Wien Austria, 2012.

2010
- ACACES Sixth International Summer School Barcelona Spain, 2010.

### Invited Talks

2025
- *Principles and Practices for Foundational Operating System Kernel Verification*. RIOS Group, National Institute of Informatics, Tokyo, Japan, November 2025.

### Service and Volunteering

2021
- SIGPLAN-M Long Term Mentor (link)

2020
- POPL'20 Artifact Evaluation Committee.

## Miscellaneous Experience (continued)

2018 PLDI'18 Student Volunteer.

2013 SEFM'13 Subreviewer.

SAS'13 Subreviewer.

2012 RV'12 Subreviewer.

## Ongoing Projects

2026 – **Reasoning principles for user trusted applications (UTAs)** (with Taro Sekiyama, CRADSEC/ROIS). Program-logic reasoning principles for trusted execution environments, modeled on OP-TEE user trusted applications: the GlobalPlatform TEE protocol for memory shared between a client and a UTA, and isolation enforced by physical memory protection, mechanized in Iris over the Sail RISC-V semantics.

2025 – **iIris: a layered Iris program logic for x86-64**, with assembly, macro, and intermediate layers composing through an inline-assembly boundary, a verified encoder/decoder and compiler, and a port of the modal abstractions to a verified xv6 kernel on RISC-V. Nearing completion.

## Research Publications

### Conference Proceedings

1. Kuru, I., & Gordon, C. S. (2025a). Modal Abstractions for Virtualizing Memory Addresses. *Proceedings of the ACM on Programming Languages*, *9*(OOPSLA2), 2338–2366. https://doi.org/10.1145/3763134
2. Kuru, I., & Gordon, C. S. (2019). A Type System for Read-Copy-Update Concurrency [Acceptance Rate 32.6% (28/86).]. *European Symposium on Programming (ESOP 2019)*. https://doi.org/10.1007/978-3-030-17184-1_4
3. Kuru, I., Matar, H. S., Cristal, A., Kestor, G., & Unsal, O. (2013). Parv: Parallelizing runtime detection and prevention of concurrency errors. In S. Qadeer & S. Tasiran (Eds.), *Runtime verification* (pp. 42–47). Springer Berlin Heidelberg.

### Thesis

1. Kuru, I. (2025). *Modal abstractions for operating system kernels* [Doctoral dissertation, Drexel University]. Drexel University. https://doi.org/10.17918/00010934

### Books and Chapters

1. Cristal, A., Ozkan, B. K., Cohen, E., Kestor, G., Kuru, I., Unsal, O., Tasiran, S., Mutluergil, S. O., & Elmas, T. (2015). *Verification tools for transactional programs* (R. Guerraoui & P. Romano, Eds.). Springer International Publishing. https://doi.org/10.1007/978-3-319-14720-8_14

### Workshop

1. Kuru, I., & Gordon, C. S. (2025c). Modal verification patterns for systems software. *Proceedings of the 13th Workshop on Programming Languages and Operating Systems (PLOS 2025), co-located with SOSP 2025*, 25–33. https://doi.org/10.1145/3764860.3768337
2. Kuru, I., Kulahcioglu Ozkan, B., Mutluergil, S. O., Tasiran, S., Elmas, T., & Cohen, E. (2014). Verifying programs under snapshot isolation and similar relaxed consistency models. *9th ACM SIGPLAN Workshop on Transactional Computing (TRANSACT 2014)*.

3. Matar, H. S., Kuru, I., Tasiran, S., & Dementiev, R. (2014). Accelerating precise race detection using commercially-available hardware transactional memory support. *5th Workshop on Determinism and Correctness in Parallel Programming (WoDet 2014)*.

### Manuscripts Under Submission

1. Kuru, I., & Gordon, C. S. (2026a). *Deferred reclamation is one discipline: A scheme-agnostic type system for memory safety, mechanized*.
2. Kuru, I., & Gordon, C. S. (2026b). *ModFs: Modal abstractions for snapshotting copy-on-write filesystems*.
3. Kuru, I., & Gordon, C. S. (2026c). *Resource morphisms for protocol indistinguishability*.
4. Liu, Z., Crawford, F. M., Kuru, I., Pichon-Pharabod, J., & Birkedal, L. (2026). *Modular and tractable verification of a realistic ticket lock on mixed-size Arm in separation logic*.

### Manuscripts In Preparation

1. Kuru, I., & Gordon, C. S. (2026d). *specNL: Protocols, resources, contexts and specification from natural language*.
2. Nishimura, K., Kuru, I., & Sekiyama, T. (2026). *A program logic for inline assembly*.

### Tech Report

1. Kuru, I., & Gordon, C. S. (2025b). Modal verification patterns for systems. `https://arxiv.org/abs/2506.01719`
2. Kuru, I., & Gordon, C. S. (2024). Modal abstractions for virtualizing memory addresses. `https://arxiv.org/abs/2307.14471`
3. Kuru, I., & Gordon, C. S. (2018). Safe deferred memory reclamation with types. `http://arxiv.org/abs/1811.11853`

## References

Available on Request